

High-performance phase change material-integrated systems for summer climate optimization

Rossella Cottone^{1,2*}, Stefano Fantucci³, Giorgia Autretto³ and Miroslav Čekon^{1,4}

¹ Slovak University of Technology, Faculty of Civil Engineering, 810 05 Bratislava, Slovakia

² KU Leuven, Department of Architecture, Campus Brussels and Ghent, 1030 Brussels, Belgium

³ Politecnico di Torino, Department of Energy, 10129 Torino, Italy

⁴ Slovak Academy of Sciences, ICA, 845 03 Bratislava, Slovakia

Abstract. Energy retrofitting and the conversion of existing buildings into dwellings are key strategies for reducing decarbonisation in the building sector and mitigating climate change. Established energy efficiency measures, such as thermal insulation, airtightness and heat recovery, have proven effective in reducing heating demand. However, as the performance of buildings improves with these approaches, they increasingly influence the overall energy balance, particularly under warm conditions. In this context, the integration of phase change materials (PCMs) into building envelope components is a promising strategy for enhancing latent thermal energy storage (LTES) and improving indoor thermal comfort in both renovations and new constructions scenarios. This study investigates the performance of an innovative integrated PCM component based on LTES principles, aimed at improving the summer performance of buildings. The proposed system consists of a 30 mm thick panel composed of paraffinic PCM impregnated in a highly porous matrix operating in synergy with a night ventilation cooling strategy. The experimental campaign was conducted on two identical test cells where the composite PCM panel was integrated into the roof of one test cell and compared to the reference without PCM. Results show that the PCM system achieved a total heat gain reduction of 23% and a 55% average decrease in peak heat flux compared with the reference configuration. The internal surface temperature decreased by up to 4.5 °C, while the operative temperature reduction reached 1.8 °C, confirming a significant improvement in indoor comfort. The observed time-lag increased between 1 h 45 min and 17 h, demonstrating the effective thermal buffering capacity of the PCM layer. Overall, these results confirm that the application of LTES panels, combined with night ventilation strategies, effectively shifts peak loads, improves indoor thermal comfort, and reduces dependence on active air conditioning systems. The versatility of the design allows for its effective use in both existing buildings with space constraints and new constructions, contributing significantly to the optimisation of energy efficiency and the thermal comfort of the indoor spaces.

Keywords. Phase Change Materials (PCM), Thermal Energy Storage, Passive-Night-Time cooling

1 Introduction

Improving the energy performance of existing buildings represents a fundamental step towards the decarbonisation of the built environment and the mitigation of climate change. In recent decades, conventional energy efficiency strategies, such as enhancing insulation levels, reducing air leakage, and recovering waste heat, have been widely implemented to lower heating energy consumption [1]. Yet, as these measures advance, new challenges emerge: buildings with high thermal performance often face increased sensitivity to overheating and cooling demands, especially in warm and transitional climates.

To address these challenges, enhancing the dynamic thermal response of building envelopes has become a key research focus. In this context, the integration of phase change materials (PCMs) offers a promising solution. By exploiting their latent thermal energy storage (LTES) capability, PCMs can absorb and release heat as ambient temperatures fluctuate, thereby contributing to indoor thermal comfort and reducing the need for active cooling [2]. This passive thermal regulation strategy is particularly relevant for both new constructions and energy retrofitting scenarios, where balancing energy efficiency with occupant comfort remains a central objective.

* Corresponding author: rossella.cottone@stuba.sk

However, to ensure effective implementation, it is essential to accurately characterize the actual thermal behaviour of PCMs under real operating conditions. Thereby, field monitoring plays a pivotal role, as it provides valuable insights into the material's performance in dynamic environments that more closely replicate real-scale applications than controlled laboratory settings.

Meng et al. [3] investigated the thermal performance of PCM-incorporated foam concrete in real buildings, demonstrating a reduction in the roof's internal surface temperature by up to 2 °C and an average decrease in heat flux of 48.5%. Thongtha et al. [4] examined the application of a PCM integrated with a fibre cement roofing slab to mitigate heat transfer through the building envelope, and reported an interior temperature reduction of 1.5 °C. In most of the aforementioned studies, PCMs were incorporated using microencapsulation or mixing techniques. Nevertheless, an alternative and straightforward integration technique involves direct impregnation [5], which exploits capillary and surface tension forces to enable effective PCM absorption within the porous structure of the host material.

Moreover, the effective application of PCMs requires careful consideration of their thermal cycling behaviour. To maximise performance, PCMs should undergo complete melting and solidification within typical daily temperature variations, ensuring efficient heat absorption and release. Studies have also shown that coupling PCM-based systems with ventilation strategies can significantly enhance their thermal charging and discharging processes [6]. Despite the growing research interest, the combined optimisation of LTES systems, ventilation strategies, and their experimental validation remains underexplored, offering a promising direction for future advancements in sustainable building design.

This study aims to evaluate the summer performance of an innovative technology that uses PCM as an LTES unit under actual operating conditions, coupled with night ventilation strategies. The system, previously developed by the author [7,8], consists of highly porous insulating panels impregnated with paraffinic PCM arranged on the sloping roof of a test room. The roof of buildings is the surface most exposed to solar radiation, making the environment below particularly subject to high summer heat loads and the risk of overheating [9,10]. The integration of PCM into the roof structure allows the accumulation of part of the incident solar radiation and environmental thermal energy. This not only attenuates temperature peaks and reduces fluctuations, helping to stabilize internal conditions, but also shifting the propagation of the heat wave towards the interior of the building.

2 Materials and Methods

This study analyzes and compares the performance of two roof systems installed in separate test rooms of the Test Roofing Innovative System (TRIS), a calibrated and

validated experimental facility developed for the thermal assessment of advanced roofing components [11].

The two test cells, located in the attic of a residential building in the metropolitan area of Turin, have an internal volume of approximately 2 m³ each and differ in the roof configuration. The first cell represents the reference (Case A), while the second (Case B) incorporates calcium silicate panels (see Figure 1) impregnated with 72 % by weight (wt.%) of Rubitherm RT27 PCM using the direct impregnation technique (PCM-B-CRT27), in order to evaluate the potential of the system when coupled with LTES.



Fig. 1. PCM-B-CRT27 installed in the Case B.

Figure 2 illustrates the layout of the temperature sensors, heat flux meters, and pyranometers used for the in-situ monitoring campaign.

The roof stratigraphies and material properties are presented in Table 1. The PCM-B-CRT27 layer, with a thickness of 30 mm, was installed near the inner surface of the room. The performance of the roof embedding the PCM panels, with a thermal transmittance value of $U=0.56 \text{ W/(m}^2\text{K)}$, was compared with that of the reference configuration without latent heat storage ($U=0.62 \text{ W/(m}^2\text{K)}$).

Figure 3 shows the specific heat capacity of the PCM-B-CRT27 composite material installed in the Case B, and measured with a Dynamic Heat Flux Meter [12]. The material exhibits a phase change temperature range between 24 °C and 29 °C, which corresponds to typical indoor overheating conditions during warm periods. The selected phase change temperature was determined

through preliminary dynamic simulations. In particular, RT27 was compared with a BioPCM characterized by a melting temperature of 29 °C, and the results demonstrated that RT27 provided a more effective utilization of latent heat within the operative temperature range of summer conditions. The phase change temperature align with the typical temperature range of building envelope applications [13]. The latent heat of fusion and solidification were measured as 110 kJ/kg and 88 kJ/kg, respectively, with melting and crystallization peaks at approximately 27 °C.

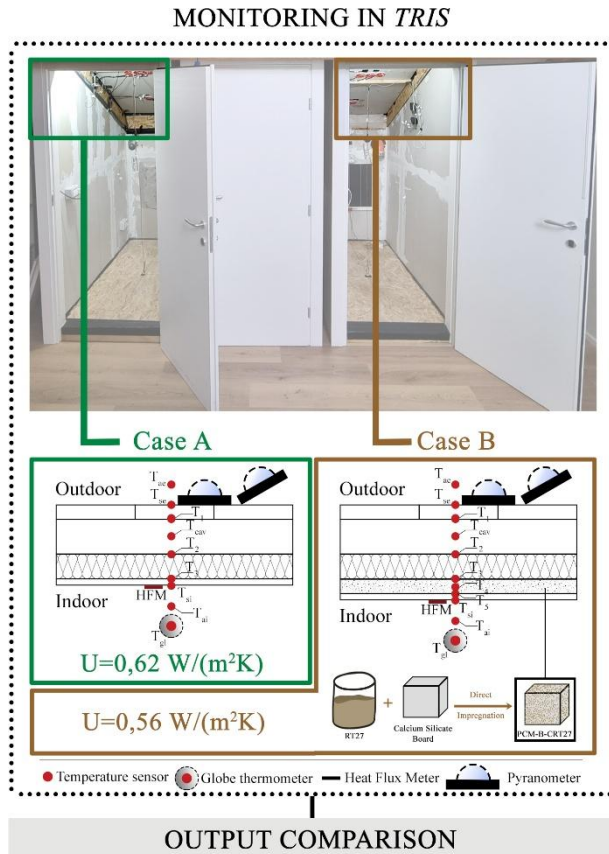


Fig. 2. Test cells and design strategies of the reference (A) and PCM-B-CRT27 (B) configurations.

Table 1. Roofing stratigraphy and materials characteristics.

Layers from outside	D [mm]	λ [W/mK]	c_p [J/kgK]	ρ [kg/m ³]
Roof Tiles	30	0.700	840	1700
Ventilated Air Gap	70	-	-	-
XPS	50	0.038	1500	33
PCM-B-CRT27	30	0.190	2400	909
Gypsum Board	12.5	0.250	1000	900

D = Thickness, λ = thermal conductivity, c_p =specific heat, ρ = density

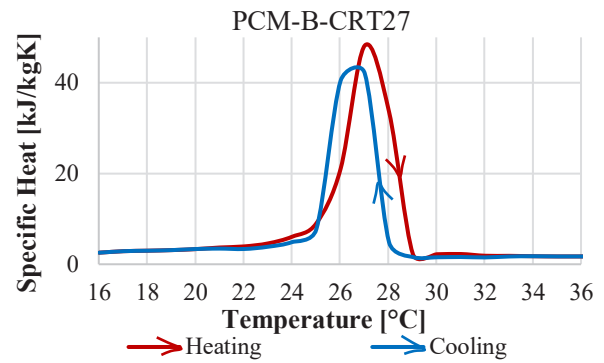


Fig. 3. Specific heat of PCM-B-CRT27 used in the Case B.

2.1 Evaluation of summer performance of the innovative system with PCM

The monitoring campaign was carried out within the TRIS experimental setup, previously calibrated and validated through experimental data. The aim of this study is to monitor and evaluate the summer performance of the innovative PCM-B-CRT27 roofing system in comparison with the reference system without PCM (see Figure 4).

In particular, a passive cooling strategy based on night ventilation (NV) was implemented to enhance the discharge of the PCM during the night. The system operated with a rate of 10 air changes per hour (ACH) from 00:00 to 08:00. Throughout the month of June, the following parameters were continuously monitored: inter-layer temperatures, outdoor air temperature, indoor air temperature, black globe temperature, solar irradiance, and heat flux through the roof assemblies. These experimental data were used to calculate key performance indicators, which enabled the assessment of the PCM-based system (Case B) and its comparison with the reference configuration without PCM (Case A).

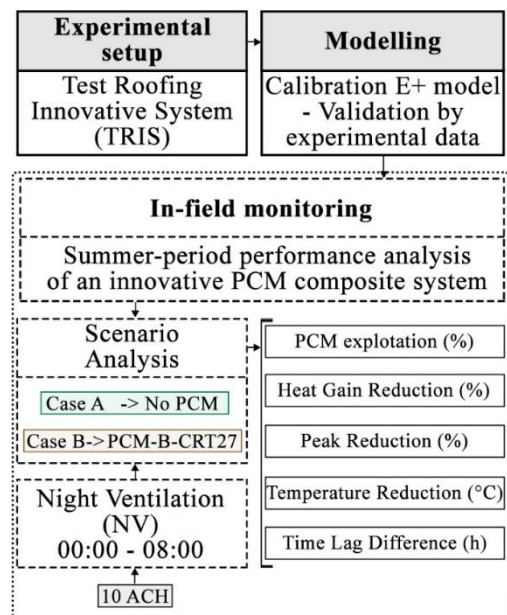


Fig. 4. Methodology workflow.

A sensor was embedded in the core of the PCM panel to record the variation of the internal temperature profile throughout the entire monitoring period. This approach enables the calculation of a performance indicator that quantifies the exploitation capability of the PCM, expressed as a percentage, according to the following equation:

$$PCM_{expl} = \frac{n^\circ \text{ of hours in the latent phase}}{n^\circ \text{ of total hours in June}} \cdot 100 [\%] \quad (1)$$

This indicator represents the fraction of time during which the PCM actively contributes to latent heat storage or release under real operating conditions.

To assess the overall thermal performance of the system, several additional key performance indicators (KPIs) were defined to quantitatively evaluate the thermal behavior of the PCM-based system compared with the reference configuration:

- i. Heat gain reduction (HGR , %), which quantifies the decrease in cumulative heat flux transmitted through the roof compared to the reference configuration:

$$HGR = \frac{Q_A - Q_B}{Q_A} \cdot 100 [\%] \quad (2)$$

where Q_A and Q_B are the total heat gains through the roof in Case A (reference) and Case B (PCM system), respectively.

- ii. Peak heat flux reduction (PR , %), measuring the reduction in the maximum heat flux during the hottest hours:

$$PR = \frac{q_{A,peak} - q_{B,peak}}{q_{A,peak}} \cdot 100 [\%] \quad (3)$$

where $q_{A,peak}$ and $q_{B,peak}$ are the peak heat fluxes measured on the inner roof surface for Cases A and B, respectively.

- iii. Surface and operative temperature reduction (ΔT , °C), which evaluates the improvement in indoor thermal conditions, expressed as the temperature difference between the reference and PCM-based systems:

$$\Delta T = T_A - T_B [^\circ C] \quad (4)$$

where T_A and T_B denote either the indoor surface temperature (T_{si}) or the operative temperature (T_{op}) for Cases A and B, respectively.

- iv. Time-lag difference ($\Delta\tau$, h), indicating the delay between the outdoor air temperature peak and the

indoor surface temperature peak, representing the PCM's thermal buffering capacity:

$$\Delta\tau = \tau_B - \tau_A [h] \quad (5)$$

where τ_A and τ_B are the time lags observed in the reference and PCM-integrated systems, respectively.

These indicators collectively provide a comprehensive evaluation of the thermal performance and energy storage potential of the innovative PCM-B-CRT27 system under real operating conditions.

3 Results and Discussions

Figure 5 shows the temperature trend in the core of the PCM-B-CRT27 panel and in the layers closest to the outer and inner surfaces, considering night ventilation active from midnight to 8 a.m. The temperatures recorded show that the temperature of the core and that of the inner surface of the panel with PCM are comparable, indicating effective thermal coupling. The highlighted area represents the latent phase range of the composite material, between 24 and 29 °C. Based on the core temperature of the PCM, the degree of PCM exploitation is estimated at 62%, calculated according to Equation 1.

However, a clear change in behavior can be observed after June 12th, when the core temperature of the PCM consistently exceeds the upper limit of the latent phase range. From this date onward, the PCM remains predominantly in the liquid state during both daytime and nighttime, despite the activation of night ventilation. As a consequence, the material progressively loses its ability to undergo effective phase change cycles, leading to a reduced latent heat storage capacity. This behavior indicates that the selected PCM melting temperature (RT27) is underestimated with respect to the actual thermal loads and boundary conditions experienced during this period. The inability of the PCM to fully solidify overnight limits its thermal buffering potential and highlights the critical importance of selecting a PCM with a melting temperature appropriately matched to the expected operating temperature range.

Figure 6 shows the daily trend of the incoming heat flow, comparing Case A (solid line) and Case B (dashed line). During the monitoring period, a marked difference between the two configurations can be observed. The reference configuration (Case A) shows more pronounced daily fluctuations, with larger peaks of heat gain and loss, while the configuration with PCM (Case B) shows a more attenuated thermal response. This behavior highlights the ability of PCM to store and release latent heat, reducing instantaneous variations in heat flow and improving the thermal stability of the building envelope.

Furthermore, analysis of the heat flow peaks shows an average reduction of 55%, with daily values ranging from 27% to 90%. These results show that the integration of

PCM not only helps to limit heat transfer through the building envelope, but also modifies the thermal dynamics of the system, attenuating daily fluctuations and reducing the intensity of peaks.

Overall, the integration of PCM into the building system allows for a significant reduction in the transmitted heat load and an improvement in the dynamic behavior of the building envelope. The combined effect of latent heat storage and release delays and reduces thermal flow peaks, promoting more stable operating conditions and potentially reducing energy demand for cooling.

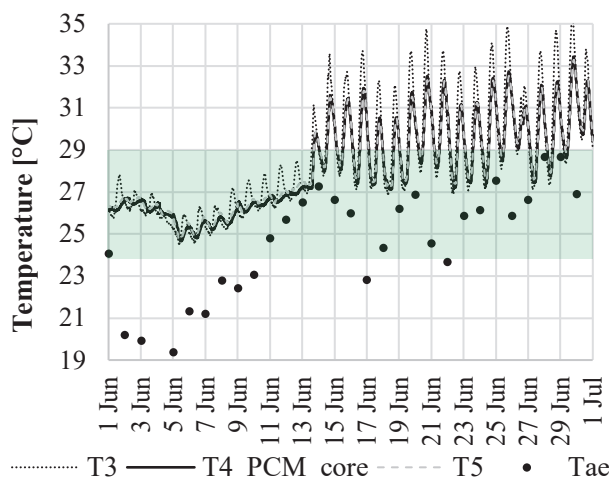


Fig. 5. Temperature profiles of PCM-B-CRT27 during June: upper surface (T3), core (T4_PCM_core), inner surface (T5), and outdoor air temperature (Tae).

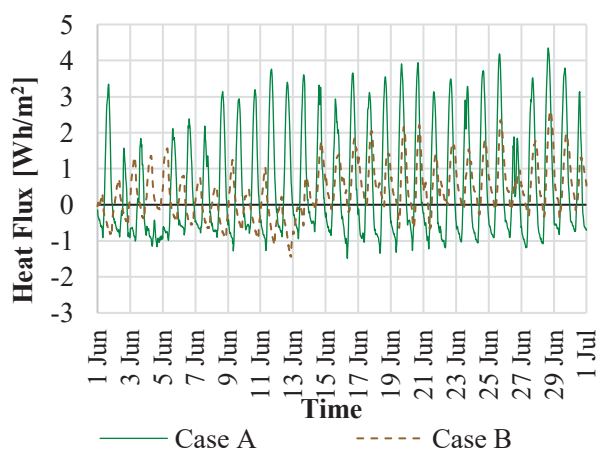


Fig. 6. Daily heat flow trend for the two configurations analyzed (Case A and Case B).

The total heat gain recorded in June is reduced from 1666 Wh/m² for Case A to 1286 Wh/m² for Case B, corresponding to a monthly reduction of 23%.

In addition, a significant reduction in internal surface temperature (Tsi) and operating temperature (Top) is observed in the PCM configuration (see Figure 7). The maximum reduction in Tsi reaches 4.5 °C, highlighting the effective thermal attenuation capacity of PCM on the internal surface exposed to the environment, while the

maximum increase, equal to +0.5 °C, is negligible, confirming that the activation of PCM does not cause discomfort during the heat release phases. Similarly, the operating temperature (Top) shows a maximum reduction of 1.8 °C and a maximum increase of +0.7 °C, indicating that PCM helps to limit heat accumulation even in the innermost layers of the system. These results confirm the effectiveness of PCM in reducing peak temperatures and stabilizing the thermal response of the building envelope, with positive effects on indoor comfort and reduction of the thermal load during the hottest hours.

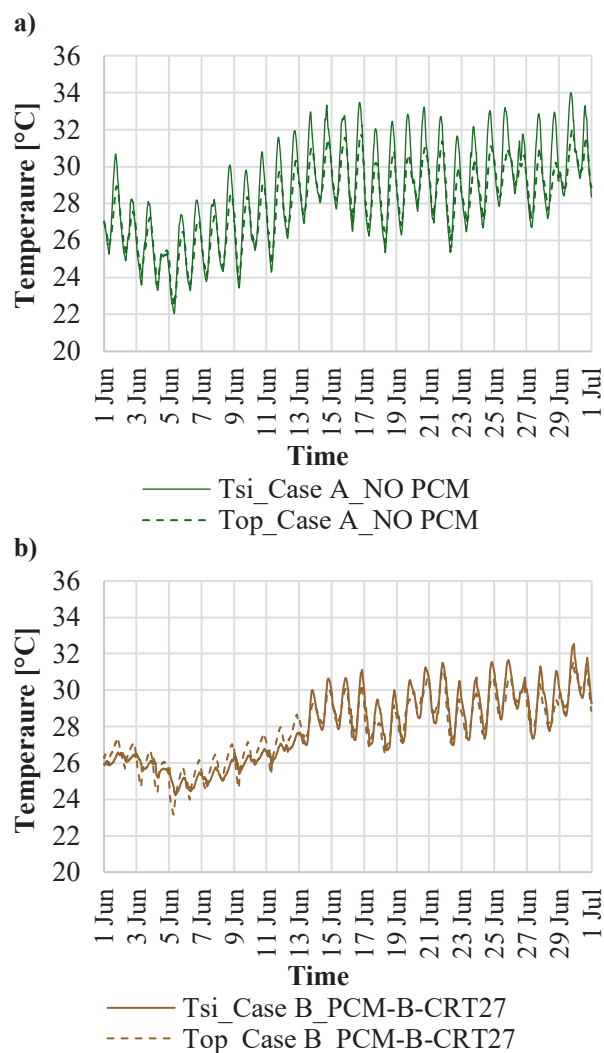


Fig. 7. Internal surface temperature of the roofs of (a) Case A and (b) Case B.

The daily variation of $\Delta\tau$ calculated using equation 5 is shown in Figure 8 together with the average temperature trend of the core of the PCM-B-CRT27. The time lag indicates the delay between the peak external temperature and the peak internal surface temperature, providing a direct indicator of the PCM's ability to dampen and delay incoming heat flows compared to the reference case (Case A). During the month of June, $\Delta\tau$ shows marked daily variability, with values generally ranging between 1 h 45

min and 17 h. An exception is observed on June 4, when the peak internal temperature occurred earlier than the external peak, suggesting a temporal reversal in the thermal response of the envelope.

In the first few days of the month, the average temperature of the PCM core is below 27 °C, indicating partial activation. Subsequently, the temperature rises until it stabilizes around 30 °C, thus operating for most of the period within or just beyond the latent phase range (see Figure 5).

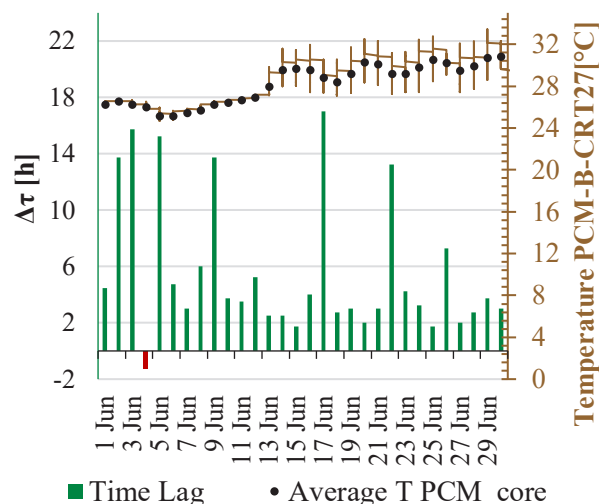


Fig. 8. Time lag variation between Case A and Case B and temperature variation of the PCM-B-CRT27 core.

Overall, the results confirm that integrating PCM into the roofing system allows for effective thermal regulation, reducing peak temperatures, mitigating daily fluctuations, and improving the dynamic stability of the building envelope, with a positive impact on both indoor comfort and thermal load. Furthermore, the system developed, analyzed in previous laboratory-scale experimental studies by the authors [7,8], has demonstrated considerable potential for application in real-world contexts.

The monitoring period, conducted during June, corresponds to the onset of sustained overheating conditions in the attic space and therefore represents a critical phase for assessing PCM activation and discharge dynamics under realistic operating conditions. The experimental results highlight that the effectiveness of the PCM-integrated roofing system is governed by the interaction between the PCM phase transition range, the diurnal temperature amplitude, and the availability of night-time cooling potential. The observed progressive reduction in latent heat utilization after mid-June indicates that when night-time temperatures remain elevated, incomplete solidification may occur, reducing the capacity for daily thermal cycling. Under milder climatic conditions or during transitional seasons characterized by larger day–night temperature differences, improved solidification is expected, leading to enhanced latent heat exploitation and greater thermal buffering. Conversely, in

persistently hot climates with limited nocturnal cooling, the system performance may progressively shift toward sensible heat storage behavior. These findings suggest that the applicability and optimal design of PCM-integrated roofing systems are climate-dependent and require alignment between the PCM melting temperature and the expected indoor operative temperature range. The experimental dataset obtained in this study therefore provides a validated reference for future modelling activities aimed at extending the assessment to different climatic contexts, seasonal conditions, and PCM configurations.

4 Conclusions

The study analyzed the performance of an innovative roofing system integrated with calcium silicate panels impregnated with PCM (PCM-B-CRT27), comparing it with a reference configuration without latent storage. The experimental results highlighted the effectiveness of PCM in reducing incoming heat flows, attenuating daily oscillations, and improving the dynamic stability of the system when the material operates within its phase change temperature range.

During the monitoring period, the PCM system achieved a monthly reduction in total heat gain of 23% compared to the reference case, accompanied by an average decrease in heat flow peaks of 55%. Significant reductions in internal surface temperature (up to 4.5 °C) and operating temperature (up to 1.8 °C) were also recorded, confirming the material's effective thermal damping capacity and improved indoor comfort conditions. Time lag analysis further confirmed the PCM-B-CRT27's ability to delay heat transfer, with $\Delta\tau$ values ranging from 1 h 45 min to 17 h, depending on operating conditions and the level of PCM activation.

At the same time, the experimental results clearly demonstrate that the effectiveness of the system is strongly dependent on the correct selection of the PCM melting temperature. During periods characterized by higher thermal loads, the core temperature of the PCM exceeded the upper limit of the phase change range, preventing complete solidification during night-time ventilation. Under these conditions, the PCM progressively lost its latent storage functionality and mainly contributed through sensible heat storage, resulting in a reduced capacity to attenuate heat flow peaks. This behavior highlights that an underestimated melting temperature limits the exploitation of latent heat and reduces the overall performance of the system, despite the presence of PCM.

The integration of PCM into the roofing system enabled effective passive control of heat flows, highlighting the system's potential for applications in real building contexts, but the results emphasize that PCM selection must be carefully tailored to the expected climatic conditions and operating temperature ranges. The choice of an appropriate melting temperature is therefore

a key design parameter to ensure full activation of the latent storage potential and to maximize the benefits of PCM-based building envelope solutions.

Future studies will focus on validating the energy model based on the experimental data collected in this study and extending the analysis to different calcium silicate-PCM configurations with different types of PCM and melting temperatures. This approach will allow optimization of the system design and a more comprehensive evaluation of PCM effectiveness across different climates and operating scenarios.

This project has received funding from the European Union's Horizon Europe research & innovation programme under the HORIZON-MSCA-2021-DN-01 grant agreement No. 101072598 – “ActaReBuild”, VEGA 2/0145/24, and the project “Chois- Characterization of Innovative and Sustainable Insulating Solu-tions”, funded within the Italian National Recovery and Resilience Plan (PNRR), Mission 4 – Education and Research- Measure M4C2 – Investment 1.1 PRIN – Research programs of national interest – project code 2022372TM9, CUP F53D23001540006.

References

- [1] IEA, Energy Efficiency 2022, (IEA, Paris, 2022). www.iea.org/reports/energy-efficiency-2022.
- [2] L. Boussaba, A. Foufa, S. Makhlof, G. Lefebvre, L. Royon, Elaboration and properties of a composite bio-based PCM for an application in building envelopes. *Constr. Build. Mater.* **185** 156–165 (2018). <https://doi.org/10.1016/j.conbuildmat.2018.07.098>.
- [3] E. Meng, J. Yang, B. Zhou, C. Wang, J. Li, Preparation and thermal performance of phase change material (PCM) foamed cement used for the roof. *J. Build. Eng.* **53**, 104579 (2022). <https://doi.org/10.1016/j.job.2022.104579>.
- [4] A. Thongtha, K. Janyoosuk, C. Mano, Integration of phase change material into fiber cement roof for reduction of heat accumulation in buildings. *ScienceAsia* **47S**, 83 (2021). <https://doi.org/10.2306/scienceasia1513-1874.2021.S017>.
- [5] C.V. Podara, I.A. Kartsonakis, C.A. Charitidis, Towards Phase Change Materials for Thermal Energy Storage: Classification, Improvements and Applications in the Building Sector. *Appl. Sci.* **11**, 1490 (2021). <https://doi.org/10.3390/app11041490>.
- [6] C. Piselli, M. Prabhakar, A. De Gracia, M. Saffari, A.L. Pisello, L.F. Cabeza, Optimal control of natural ventilation as passive cooling strategy for improving the energy performance of building envelope with PCM integration. *Renew. Energy* **162**, 171–181 (2020). <https://doi.org/10.1016/j.renene.2020.07.043>.
- [7] R. Cottone, M. Čekon, C. Glorieux, S. Fantucci, J. Čurpek, R. Slávik, M. Rychtáriková, Investigation of Phase Change Material Impregnation in a Highly Porous Structure, in *Multiphysics and Multiscale Building Physics*, U. Berardi (Ed.), (Springer Nature Singapore, Singapore, 2025) **552**, pp. 455–460. https://doi.org/10.1007/978-981-97-8305-2_64.
- [8] R. Cottone, M. Čekon, R. Slávik, C. Glorieux, S. Fantucci, P. Matiasovsky, M. Rychtáriková, Innovative PCM-integrated calcium silicate panels for thermal regulation in building envelopes. *Results Eng.* **29** (2026) 109252. <https://doi.org/10.1016/j.rineng.2026.109252>.
- [9] H.-Y. Liu, N. Skandalos, L. Braslina, V. Kapsalis, D. Karamanis, Integrating Solar Energy and Nature-Based Solutions for Climate-Neutral Urban Environments. *Solar* **3**, 382–415 (2023). <https://doi.org/10.3390/solar3030022>.
- [10] H. Elarga, S. Fantucci, V. Serra, R. Zecchin, E. Benini, Experimental and numerical analyses on thermal performance of different typologies of PCMs integrated in the roof space. *Energy Build.* **150**, 546–557 (2017). <https://doi.org/10.1016/j.enbuild.2017.06.038>.
- [11] S. Fantucci, G. Autretto, E. Fenoglio, E. Sassaroli, M. Perino, Laboratory Assessment and In-Field Monitoring of Macro-Encapsulated Phase Change Materials for Building Envelope Applications. *Appl. Sci.* **12**, 2054 (2022). <https://doi.org/10.3390/app12042054>.
- [12] ASTM C1784, Standard test method for using a heat flow meter apparatus for measuring thermal storage properties of phase change materials and products. ASTM International (2014).
- [13] L.F. Cabeza, A. Castell, C. Barreneche, A. de Gracia, A.I. Fernández, Materials used as PCM in thermal energy storage in buildings: A review. *Renew. Sustain. Energy Rev.* **15** (2011) 1675–1695. <https://doi.org/10.1016/j.rser.2010.11.018>.